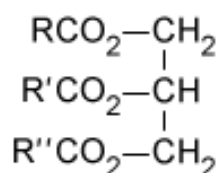


Which statement correctly describes the hydrolysis of 1 mole of the compound shown below with 3 equivalents of NaOH? The basic hydrolysis of this molecule:



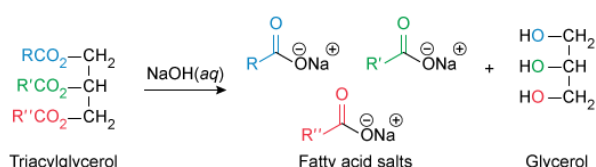
- ☐ A. is a reduction reaction. [5%]
- ☐ B. generates 3 moles of one fatty acid salt and 1 mole of glycerol. [22%]
- ☐ C. is an $\text{S}_{\text{N}}2$ reaction. [3%]
- ✓ ☒ D. generates 1 mole for each of three different fatty acid salts and 1 mole of glycerol. [69%]

Correct

69% answered correctly

Explanation:

Basic hydrolysis of a triacylglycerol



When a triacylglycerol has three different hydrocarbon chains, **three different fatty acid salts** and glycerol are released during hydrolysis

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Triacylglycerols are lipids made up of three **fatty acyl groups** and a molecule of **glycerol**. Each fatty acyl group is bonded to glycerol through an **ester** linkage. Hydrolysis of the triacylglycerol can occur under acidic or basic conditions and breaks the ester bonds, releasing glycerol and three fatty acids under acidic conditions or three fatty acid salts under basic conditions.

Base-mediated ester hydrolysis is known as **saponification** and begins by the addition of hydroxide to the carbonyl, creating a tetrahedral intermediate. The fatty acid salt is released from the glycerol molecule when the carbonyl is reformed. The **number of unique fatty acid salts** released through hydrolysis **depends on** the **composition** of the **hydrocarbon tails**:

- All hydrocarbon tails are the same—only one unique fatty acid salt is released
- Two hydrocarbon tails are the same and one is different—two unique fatty acid salts are released

- All the hydrocarbon tails are different—three unique fatty acid salts are released

The hydrocarbon tails on the triacylglycerol in this question are represented by R, R', and R''. Because each R group is different, the triacylglycerol has **three different hydrocarbon tails**. Therefore, **three unique fatty acid salts** will be released **along with glycerol** upon basic hydrolysis of the triacylglycerol.

(Choice A) In the basic hydrolysis of a triacylglycerol, no change in the number of C–O bonds occurs. Therefore, the reaction is *not* a **reduction**.

(Choice B) If *all* of the R groups (hydrocarbon tails) on the three fatty acyl groups were the *identical*, then basic hydrolysis of the triacylglycerol in this question would give 3 moles of the *same* fatty acid salt.

(Choice C) **S_N2 reactions** are a concerted process. Because the hydrolysis of a triacylglycerol occurs in more than one step as a **nucleophilic acyl substitution**, it *cannot* be an S_N2 reaction.

Educational objective:

Triacylglycerols are composed of three fatty acyl groups bonded to glycerol through ester linkages. Basic hydrolysis of the ester linkage releases three fatty acid salts and glycerol. The number of unique fatty acid salts released depends on the composition of the hydrocarbon tails on the fatty acyl groups.

Time spent:0 sec

Subject:

Organic Chemistry

QID:404060

System:

Functional Groups and Their Reactions